

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 10 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

Class: Class 10

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking Concepts, Internet and Web Technologies

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** b - Transmission Control Protocol/Internet Protocol
 - **Explanation:** TCP/IP stands for Transmission Control Protocol/Internet Protocol, which is the standard communication protocol suite for the Internet.
2. **Answer:** c - Transmission Control Protocol/Internet Protocol
 - **Explanation:** The full form is Transmission Control Protocol/Internet Protocol. It governs how data is transmitted across networks.
3. **Answer:** c - Router
 - **Explanation:** A router connects multiple networks and directs data packets between them based on IP addresses.
4. **Answer:** b - Educational institutions
 - **Explanation:** The .edu top-level domain is reserved for educational institutions such as universities and schools.
5. **Answer:** a - 2048
 - **Explanation:** The practical maximum length of a URL is approximately 2048 characters as per standard browser limitations.

Section II: Short Answer (5 x 2 = 10 marks)

1. **Differentiate between LAN and WAN**
 - **Answer:** LAN (Local Area Network) covers a small geographical area like a school or office building. WAN (Wide Area Network) covers a large geographical area like a city or country. LAN examples: school computer lab network, office home network. WAN examples: Internet, bank network connecting branches across cities.
 - **Marking:** 1 mark for each correct differentiation point, 0.5 marks for each example.
2. **Explain the client-server model**
 - **Answer:** In the client-server model, clients (computers) request services from a central server. The server processes requests and returns responses. Diagram should show multiple clients connected to a central server. Example: When you open a website, your browser (client) requests a webpage from the web server.
 - **Marking:** 1 mark for diagram, 1 mark for explanation with example.
3. **Role of DNS server and domain name resolution**
 - **Answer:** DNS (Domain Name System) translates human-readable domain names (www.example.com) into IP addresses. When you type a URL, the browser contacts a DNS server which looks up the IP address. The DNS server checks its cache, then queries root servers, TLD servers, and authoritative servers to resolve the name.
 - **Marking:** 0.5 marks for DNS purpose, 1 mark for resolution process, 0.5 marks for example.

4. Compare HTTP and HTTPS

- **Answer:** HTTP (HyperText Transfer Protocol) transmits data in plain text. HTTPS (HyperText Transfer Protocol Secure) encrypts data using SSL/TLS. HTTPS is preferred for banking because it ensures data confidentiality, integrity, and authentication. It prevents eavesdropping and man-in-the-middle attacks.
- **Marking:** 1 mark for comparison, 1 mark for banking preference explanation.

5. What is cloud computing? List service models

- **Answer:** Cloud computing delivers computing services (servers, storage, databases, networking) over the Internet on a pay-as-you-go basis. Service models: IaaS (Infrastructure as a Service) provides virtual machines and storage; PaaS (Platform as a Service) provides development platforms; SaaS (Software as a Service) provides ready-to-use applications.
- **Marking:** 1 mark for definition, 0.5 marks for each correct service model.

Section III: Long Answer (5 x 1 = 5 marks)

1. TCP/IP protocol suite

- **Answer:** The TCP/IP model has 4 layers: (1) Application Layer - HTTP, FTP, SMTP, DNS protocols handle user applications; (2) Transport Layer - TCP and UDP ensure reliable/unreliable data delivery between hosts; (3) Internet Layer - IP protocol handles addressing and routing of packets; (4) Network Access Layer - Ethernet and Wi-Fi handle physical transmission. Diagram should show all 4 layers with protocols at each level.
- **Marking:** 1 mark for correct diagram, 1 mark for each layer description with protocol (up to 3 marks).

Part B: Hands-on Practical Lab Task (5 marks)

Program: CSV Student Data Processing

```
import csv

# Read CSV and process student data
students = []
with open('students.csv', 'r') as f:
    reader = csv.DictReader(f)
    for row in reader:
        name = row['Name']
        roll = row['Roll']
        marks = [int(row['Sub1']), int(row['Sub2']), int(row['Sub3'])]
        total = sum(marks)
        percentage = total / 3

        if percentage >= 90:
            grade = 'A'
        elif percentage >= 75:
            grade = 'B'
        elif percentage >= 60:
            grade = 'C'
        else:
            grade = 'D'

        students.append({
            'Name': name, 'Roll': roll, 'Total': total,
            'Percentage': percentage, 'Grade': grade
        })
```

```
# Write results to new file
with open('results.csv', 'w', newline='') as f:
    writer = csv.DictWriter(f, fieldnames=['Name', 'Roll', 'Total', 'Percentage',
    'Grade'])
    writer.writeheader()
    writer.writerows(students)

# Find topper
topper = max(students, key=lambda x: x['Percentage'])
print(f"Class Topper: {topper['Name']} with {topper['Percentage']:.2f}%")
```

- **Marking:** 1 mark for CSV reading, 1 mark for grade calculation, 1 mark for file writing, 1 mark for topper finding, 1 mark for code structure.

Part C: Notebook Maintenance (5 marks)

1. Neatness and organisation of notebook (2 marks)
2. Completion of all assigned classwork and homework (2 marks)
3. Proper headings, dates, and indexing (1 mark)